Smart Housing Scout

Rent with evidence, not claims.

ASSIGNMENT 9: SOFTWARE PROJECT

A web app for students who rent in Pittsburgh without seeing the home first. It ranks homes by real door-to-campus commute time, true monthly cost, and verified bathroom count, instead of what the listing says.

QUESTION 1

Functional decomposition (3 points)

The digital experience: an international student choosing an apartment in Pittsburgh, near Carnegie Mellon, before arriving in the city.

User objective (1 point)

The users are college students from overseas who need to find accommodation before they arrive. They want to see what a place really looks like, and to know three things a listing may not tell them honestly: how long it really takes to get from the door to campus, not just how far it is (the Commute Truth-Teller); whether “2 baths” means two full bathrooms or one bathroom and a toilet-only washroom; and what the home will really cost each month with utilities. This matters because they need actual numbers to plan: a student with a $700 monthly budget and a $500 rent needs to know that utilities will stay under $200.

Primary functions and system classification (2 points)

#	FUNCTION	DONE BY	FRONT OR BACK END	WHERE IT MEETS THE REAL WORLD
1	Choose the campus and the time class starts ("Arrive by")	User	Front end	The student's real class schedule
2	Search by neighborhood and set filters: budget, beds, baths, max commute, utilities	User	Front end	Keyboard, mouse, or touchscreen
3	Turn "8:45" into the next weekday at 8:45 Pittsburgh time	Program	Back end	The real calendar and the Eastern time zone

#	FUNCTION	DONE BY	FRONT OR BACK END	WHERE IT MEETS THE REAL WORLD
4	Get a transit route from each home to the campus building	Program	Back end (server)	Google Routes API, which uses real bus schedules. Sample model when no key is set
5	Split the route into walk, wait, ride, transfer, final walk, and early arrival; compute total minutes	Program	Back end (logic)	Real stop locations and departure times
6	Estimate true monthly cost: rent plus electricity, gas, water, internet, insurance	Program	Back end (logic, runs in browser)	Square footage, heating type, season
7	Compare listed baths with verified full and half baths; raise flags	Program	Back end (logic)	Tenant photos and floor plans
8	Show cards, map pins, Journey Bars, photos, tables, and plain-language findings	Program	Front end	The screen; OpenStreetMap tiles
9	Save homes and pick 2 to 4 to compare	User	Front end	Stored in the browser (localStorage)
10	Decide which home to rent and contact the landlord	User	Outside the software	A real lease and a real move

Note: the commute logic runs on a small Node server so the Google API key never reaches the browser. The cost, bathroom, and comparison rules run in the browser, but they are still back-end logic: they apply rules, while the page only displays results.

QUESTION 2A

Project description and architecture (6 points)

Project description

Smart Housing Scout is a commute truth teller and lease time machine. It lets you know exactly what your house is like from overseas and if it truly fits your budget. It shows your true daily commute to campus, based on bus schedules, along with real stats and verified reviews from past renters.

Target user

The users are international college students moving to Pittsburgh who need to secure accommodation before they arrive.

Requirements

The app must rank homes by real door to campus commute time rather than distance. It needs to break the trip down into a visual Journey Bar, calculate the true monthly cost including utilities, flag

mismatches in bathroom counts, let users compare options side by side, and work offline using sample data when no Google API key is set, and show a clear error message if the server is down.

#	REQUIREMENT	MET?
R1	Rank homes by door-to-campus commute time by default, with distance as secondary text	**Yes**
R2	Break each trip into walk, wait, ride, transfer, final walk, and early arrival, and draw it as a proportional Journey Bar	**Yes**
R3	Say in plain words when a closer home is slower ("Closer than Penn Ave, but 21 min slower")	**Yes**
R4	Show the true monthly cost with a toggle per utility and a season choice; label every estimate	**Yes**
R5	Flag a mismatch between listed and verified baths	**Yes**
R6	Show unstaged tenant photos, a before/after slider, and verified tenant reviews	**Yes**
R7	Compare 2 to 4 homes and highlight the best value in each row	**Yes**
R8	Keep the Google API key on the server; work fully offline with sample commutes when no key is set	**Yes, sample mode tested. Google mode not tested (no key)**
R9	Loading, empty, and error states; keyboard focus; works on a phone	**Yes**

Inputs and outputs

Inputs include the target campus, a specific arrive by time, search text, and filters for budget, beds, and baths. Outputs provide the step by step commute breakdown, estimated true monthly utility cost, verified tenant ratings, and plain language mismatch flags.

INPUTS	OUTPUTS
Campus (CMU, Pitt, Duquesne)	Commute: total minutes, each segment, bus lines, transfers, leave and arrive times
Arrive-by time (7:00 to 11:00 AM)	
Search text	True monthly cost and each utility estimate ($)
Filters: max budget, min beds, min verified full baths, max commute, utilities included	Bath check: listed vs verified, mismatch flag
	Lease flexibility score (0 to 4)
Sort order	Tenant rating (1 to 5)
Per-home: season, "included in rent" toggles	Flags and plain-language findings

Fixed values

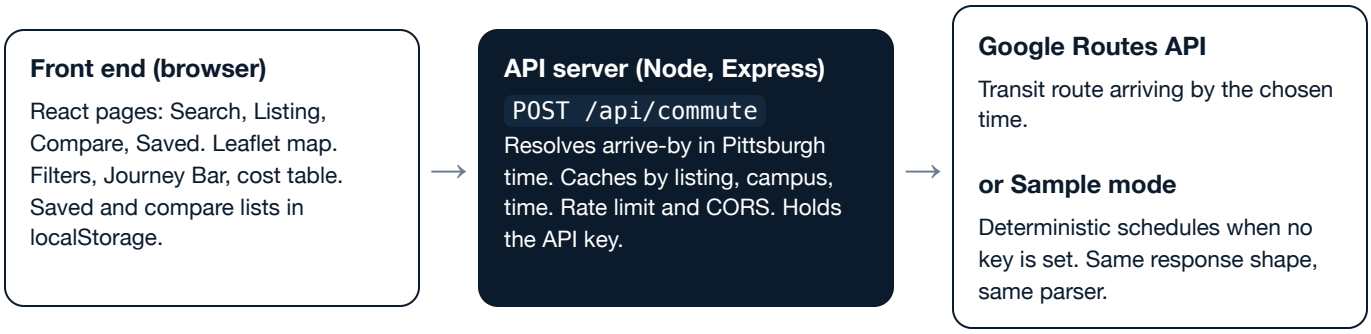
ITEM	VALUE
Campus buildings (where trips end)	CMU Gates Hillman Center; Pitt Cathedral of Learning; Duquesne Old Main
Time zone	America/New_York, computed on the server

ITEM	VALUE
Electricity ($ per sq ft per month)	Winter 0.085, summer 0.12, other months 0.07; electric heat adds 0.20 in winter
Gas ($ per sq ft per month)	Winter 0.17, summer 0.025, other months 0.06
Water and sewer; internet; insurance	$22 + $16 per bedroom; $65; $15
Building efficiency multiplier	Efficient 0.85, average 1.0, drafty 1.4 (on heating and cooling)
"Meaningful" commute difference	3 minutes or more

How it works (logic and math)

If you select CMU, the app finds the actual bus going to CMU from a specific stop rather than a random one, aligning with transit APIs for accurate schedules (sample schedules until a Google API key is added). For the commute math, the total time equals your arrive by time minus your departure time, meaning early arrival is accurately counted as commute time. True cost is calculated by adding rent and estimated utilities based on square footage, season, and heating type. The Bathroom Detective compares the listed count against verified data where a half bath equals 0.5. It raises a red flag if the numbers do not match.

Architecture



Shared code (`shared/`) is used by both sides: types, sample listings, the route parser, the cost model, and the comparison rules.

Success metrics

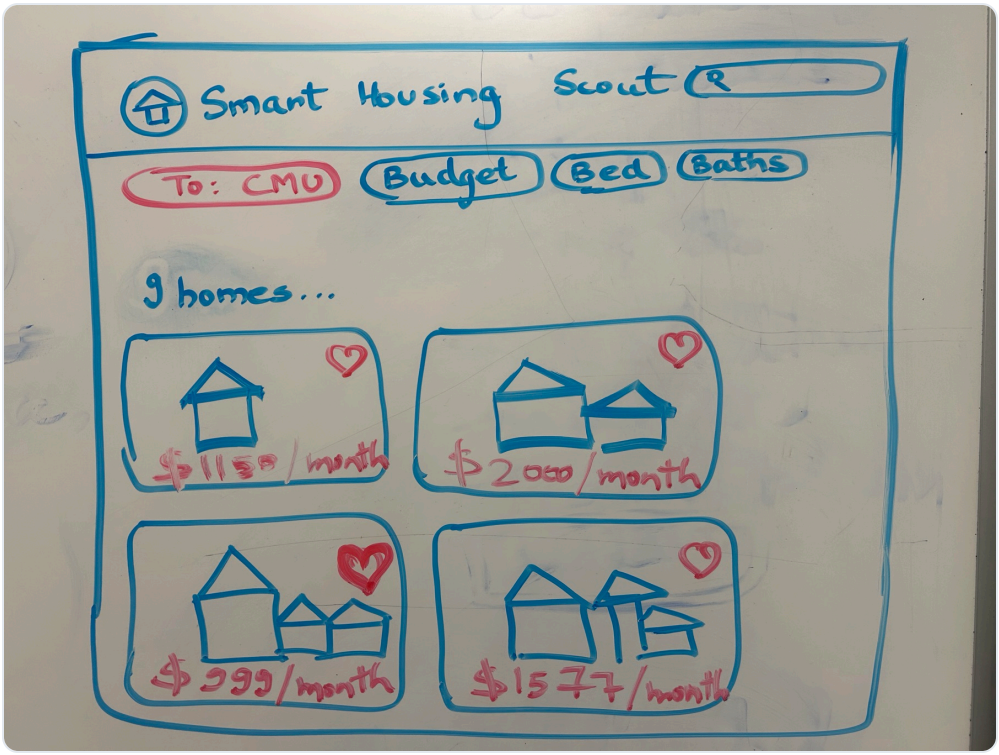
Success looks like students actually using the app to finalize houses without depending on Zillow, Google Maps, or video calling, completely removing friction at all layers. Technically, success means all five test traps trigger correctly in the frontend development, commute segments perfectly add up to the total time, and the cost table perfectly matches manual calculations.

Testing plan

The testing plan relies on five built in traps to catch edge cases: a home that is physically close but has a slow commute, a farther home with a fast direct bus, a listing claiming 2 baths that is verified as 1.5, a cheap rent with hidden high utilities, and a home with great reviews but the longest commute.

TRAP	HOME	EXPECTED RESULT (CMU, ARRIVE 8:45 AM)	RESULT
Closest home, slow commute	Semple St, 0.8 mi	39 min: 75 bus then 61C, arrives 14 min early. Not the fastest	**Pass**
Farther home, fast direct bus	Penn Ave, 2.3 mi	18 min on the 71D, no transfer. Fastest of 12	**Pass**
Listed 2 baths, really 1.5	Ellsworth Ave	Red mismatch flag: 1 full + 1 half	**Pass**
Cheap rent, high utilities	Greenfield Ave	Rent $950, true cost $1,405 (average), $1,560 in winter	**Pass**
Great reviews, long commute	Beechwood Blvd	Rating 4.8 of 5, commute 43 min (slowest)	**Pass**

Interface sketch



Hand-drawn sketch of the search page, made before building.

QUESTION 2B

Working demonstration code (6 points)

1. Initial prompt

Submitted to Claude Code in one message. The full text is below, unchanged.

> You are a senior product designer and full-stack engineer. Build "Smart Housing Scout", a rental search web app for students who rent sight-unseen. It should have the structure of Zillow (map and list side by side, filters, saved homes) and the craft of Airbnb (clean, calm, image-led, every detail finished). Its difference: every claim in a listing is checked and shown as evidence.
>
> PRODUCT

- Name: Smart Housing Scout. Tagline: "Rent with evidence, not claims."
- Users: students and international students relocating to a city they have never visited. Default city: Pittsburgh. Default campus: Carnegie Mellon University. The campus selector includes CMU, University of Pittsburgh, and Duquesne University.
- Core idea: listings say "5 minute walk to campus", "2 bath", "utilities included", and show staged photos. Scout replaces each claim with something verified.

THREE SIGNATURE FEATURES
1. Commute Truth-Teller (the MVP; build it first and best)
 - Compares homes by true door-to-campus commute time, not distance.
 - For each listing, get a transit route from the listing's coordinates to the selected campus building, arriving by a chosen time (default 8:45 AM on the next weekday).
 - Break the trip into four parts: walk to stop, wait, ride, walk to building. Wait = transit departure time minus (trip start + walk-to-stop time). Show line names, number of transfers, and total minutes.
 - Signature visual: a proportional "Journey Bar", four colored segments scaled to minutes (walk teal, wait amber, ride blue, final walk light teal), with minutes labeled.
 - Default sort is "Fastest commute". Show distance as secondary text. Where the closest home is not the fastest, say so in plain words, for example "Closer than Maple St, but 11 min slower".
2. Lease Time Machine (shows what a listing hides), on the detail page as three tabs:
 - Real photos: unstaged, multi-angle room photos uploaded by past tenants, each with room label, angle, and date. Include a before/after slider that compares the listing photo with the unstaged photo of the same room. State clearly that photos are unstaged and not AI-edited.
 - True monthly cost: rent plus estimated electricity, gas, water and sewer, internet, and renters insurance, as a stacked bar and a table. Headline: "Listed $1,450. True monthly cost $1,687." A toggle for each utility lets the user mark it "included in rent". Estimates come from sample factors per square foot and season, labeled "estimate".
 - Past tenants: overall rating plus category ratings (landlord responsiveness, maintenance, noise, winter heating), and two or three short reviews with tenancy dates and a "Verified tenant" badge.
3. Bathroom Detective
 - Compare the listed bath count with the verified breakdown (full and half baths). If they differ, show a red mismatch flag with a one line explanation.
Also include a Lease terms panel (length options, deposit, early termination, sublease allowed, notice period) and a short accordion of GENERAL tenant-rights information, labeled "general information, not legal advice". Do not invent specific laws or statute numbers.

SCREENS
1. Search (home): sticky top bar with logo, one rounded search field (neighborhood, address, or campus), an "Arrive by" time picker, and filter pills (budget, beds, baths, max commute minutes, utilities). Below it, a split view. Left: scrollable result cards. Right: a map with price pins. Hovering a card highlights its pin, and clicking a pin scrolls to its card. On mobile: a map and list toggle with a bottom sheet.
2. Listing card: image carousel with dots and a heart (save), then price, beds, baths, square feet, address, a commute chip ("31 min door to campus") with a mini Journey Bar, and small badges ("Baths verified", "Unstaged photos", "True cost +$237").
3. Listing detail: Airbnb-style gallery (one large image and four small, with "Show all photos"), a sticky right panel with the true monthly cost and Save, Compare, and Contact buttons, and sections for Commute Truth-Teller, Lease Time Machine, Bathroom Detective, Lease terms, and neighborhood.
4. Compare (2 to 4 homes): a pivot table with homes as columns and rows for true commute (Journey Bars), true monthly cost, baths (listed vs verified), lease flexibility, tenant rating, and flag count. Highlight the best value in each row.
5. Saved homes.

DESIGN SYSTEM: Zillow's structure, Airbnb's finish
- Light, airy listing surfaces: white (#FFFFFF) cards over mist (#F6F8FA). Deep midnight navy (#0B1B2B) for the header, nav, and the evidence panels. Evidence values (minutes, dollars, counts) use tabular numerals or a monospace face, so verified data looks different from listing claims.
- Colors: primary action Scout green #0E9F6E, warning and hidden cost amber #F5A524, mismatch red #E5484D, transit blue #2F6FED, hairline borders #E3E8EE. Do not copy Zillow's blue or Airbnb's red.
- Type: Inter for UI, tabular numerals for numbers, one display weight for headings. Scale 12 / 14 / 16 / 20 / 28 / 40, line height 1.4 to 1.5, line length under 70 characters.
- Layout: 8 pixel grid. 16 to 20 pixel radius on images and cards, 10 to 12 on inputs, fully rounded pills and buttons. Soft layered shadows only on hover and on floating elements. Generous whitespace. Photos are the hero of every card.
- Motion: small and purposeful only. Card lift on hover, heart pop on save, carousel slide, Journey Bar segments growing in once, skeleton loaders. Respect reduced-motion settings.
- States: loading skeletons, empty states with a clear next step, error states that say what happened and how to fix it. Visible keyboard focus. WCAG AA contrast. Fully responsive.
- Copy: plain and specific. No marketing filler. No lorem ipsum.

DATA
- Use clearly labeled sample data: 12 fictional listings across Shadyside, Squirrel Hill, Oakland, Bloomfield, Friendship, Point Breeze, and Greenfield, with realistic coordinates, rents ($900 to $2,400), unit photos, utilities, reviews, and lease terms. Use royalty-free placeholder images with stable IDs.
- Build in these test traps: one home that is closest but has a slow commute, one farther home with a fast direct bus, one listing that says 2 baths but is verified 1 full and 1 half, one cheap rent with high utilities, and one with excellent tenant reviews but a long commute.
- Define TypeScript types for Listing, CommuteResult (walkToStop, wait, ride, walkToBuilding, transfers, lines, totalMinutes), CostBreakdown, TenantReview, and BathroomCheck.

TECH
- React with TypeScript, Vite, Tailwind CSS, and React Router. Leaflet with OpenStreetMap tiles for the map, so no map key is needed.
- A small Node and Express backend that proxies Google's Routes API for transit: POST https://routes.googleapis.com/directions/v2:computeRoutes with travelMode TRANSIT and arrivalTime, using the headers X-Goog-Api-Key and X-Goog-FieldMask. Request only the fields needed (routes.legs.steps.transitDetails, step durations, start and end times, total duration). Keep the API key on the server in an environment variable, never in the front end. Provide a .env.example.
- Endpoint: POST /api/commute with origin, destination, and arriveBy. Cache results in memory by listing, campus, and time. Add a simple rate limit and CORS.
- Mock mode: if no API key is set, return deterministic sample commutes and show a small "Sample commute data" label, so the app works fully offline.
- Write unit tests for the function that turns a Routes API response into the four journey parts, using a fixture JSON.

OUT OF SCOPE
User accounts, payments, real listing scraping, and the Zillow API.

HOW TO WORK
Build in this order: (1) design tokens and layout shell, (2) sample data and types, (3) search page with map and cards, (4) Commute Truth-Teller with backend and mock mode, (5) listing detail with the Lease Time Machine and Bathroom Detective, (6) compare page and saved homes, (7) polish and accessibility. Do not ask me questions. Make

reasonable assumptions and list them at the end. Do not claim anything works until you have run it, and tell me exactly how to run it and where to add the Google API key.

REVIEW NOTES SENT WITH THE PROMPT (SUMMARY)

The same message included a review of the prompt. Claude applied each point:

- **Early arrival:** with "arrive by 8:45", the real loss is landing at 8:22. Added as a fifth segment; total = arrive-by – departure.
- **Transfers:** a trip with a transfer has two rides and two waits. The Journey Bar shows the trip as an ordered sequence.
- **Routes API field names:** check the docs. Claude confirmed that steps use `staticDuration` and that a route has no departure time.
- **Campus building** and **time zone** (America/New_York on the server) defined.
- Utility factors written out, photo pairs labeled illustrative, localStorage for saved homes, "best" defined per Compare row, tests beyond the parser, commit after each stage.

2. The finished interface

All screenshots: captured by Claude with headless Chrome from the running app, Nov 9, 2026, sample commute data.

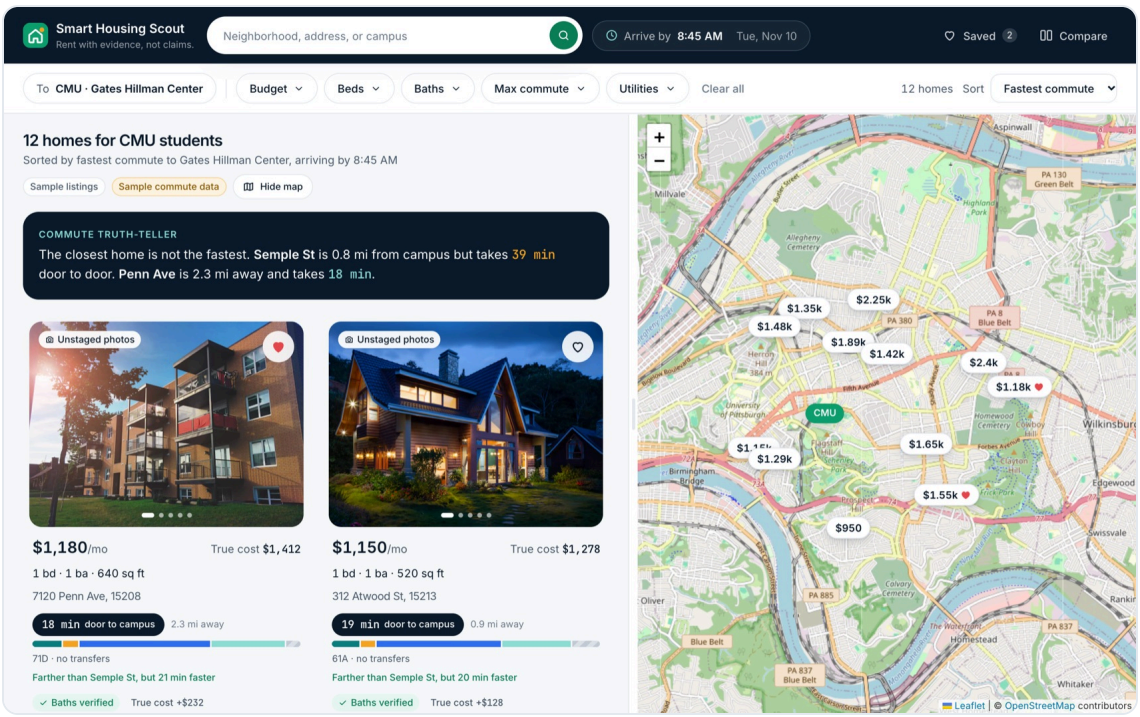


Figure 1. Search page. Cards sorted by fastest commute, map with price pins, and the finding "The closest home is not the fastest". The bar between list and map resizes the map; "Hide map" removes it.

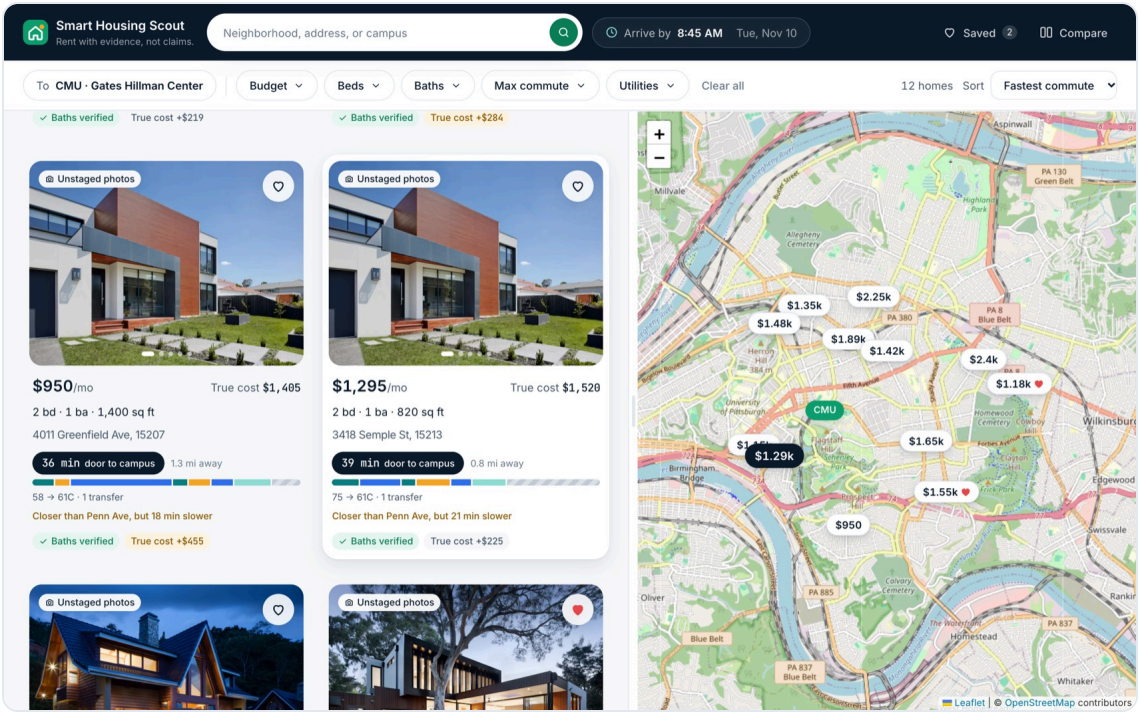


Figure 2. Hovering the Semple St card highlights its pin on the map ($1.29k, dark). Each card shows the mini Journey Bar and a finding such as "Closer than Penn Ave, but 21 min slower".

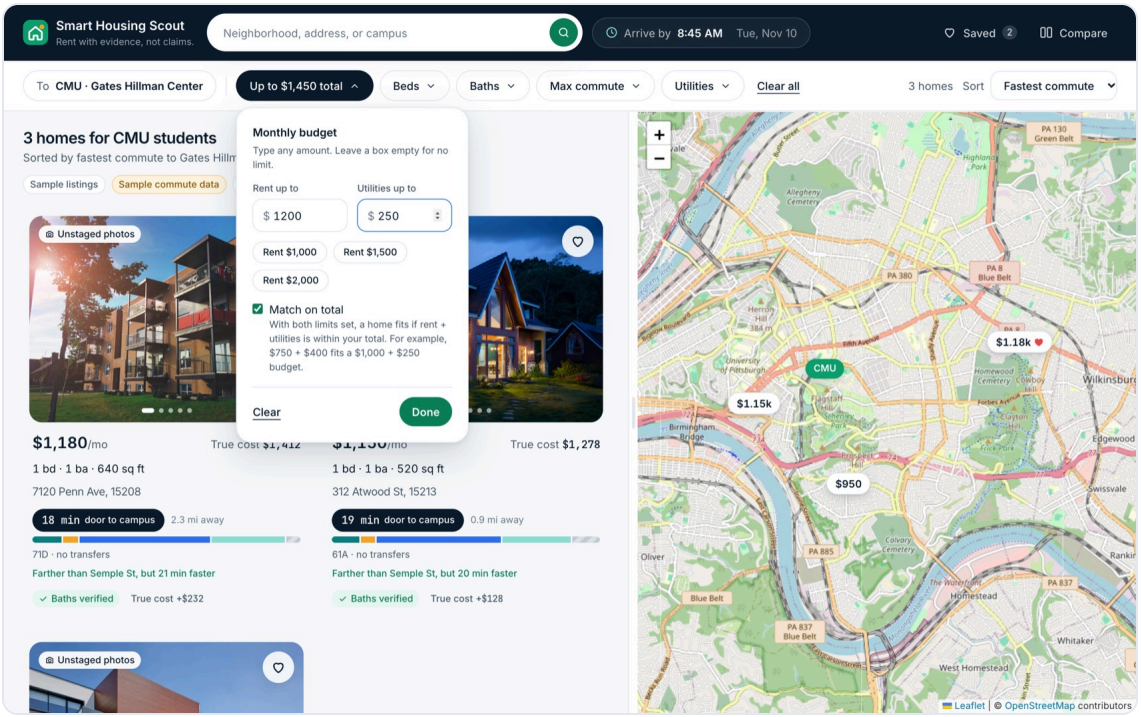


Figure 3. Budget filter with typed limits: rent $1,200 and utilities $250. With "Match on total", 3 homes fit within $1,450 in total. Every filter has its own Clear, and "Clear all" resets them.

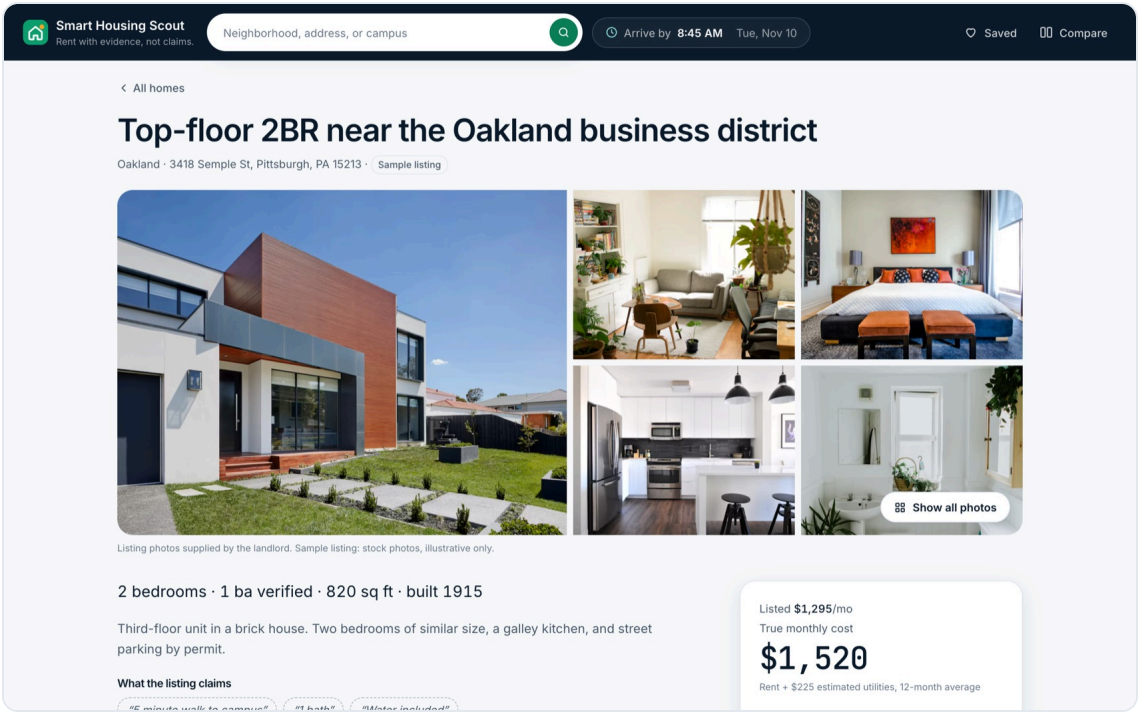


Figure 4. Listing detail: gallery (one large, four small), the listing's own claims, Scout's flag, and the sticky cost panel.

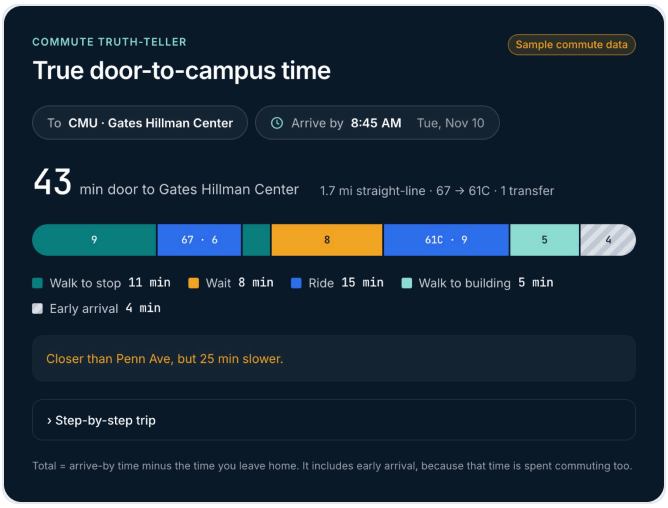


Figure 6. The best-reviewed home (Beechwood Blvd) has the longest commute: 43 min, infrequent 67 bus plus a transfer.

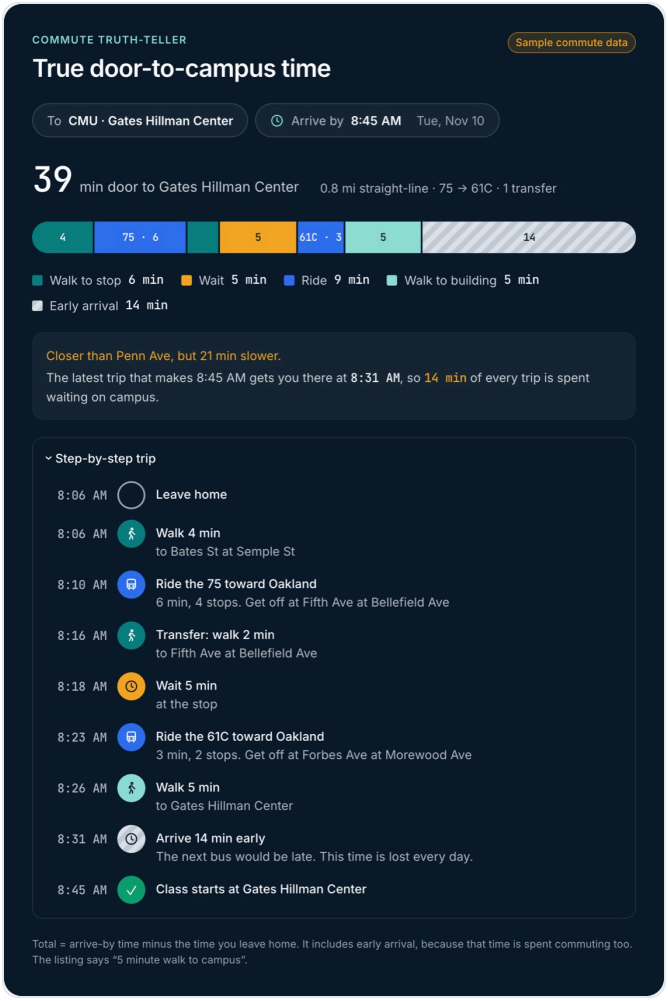


Figure 5. Commute Truth-Teller for the closest home. The listing says "5 minute walk to campus"; the real trip is 39 min with a transfer and 14 min of early arrival.

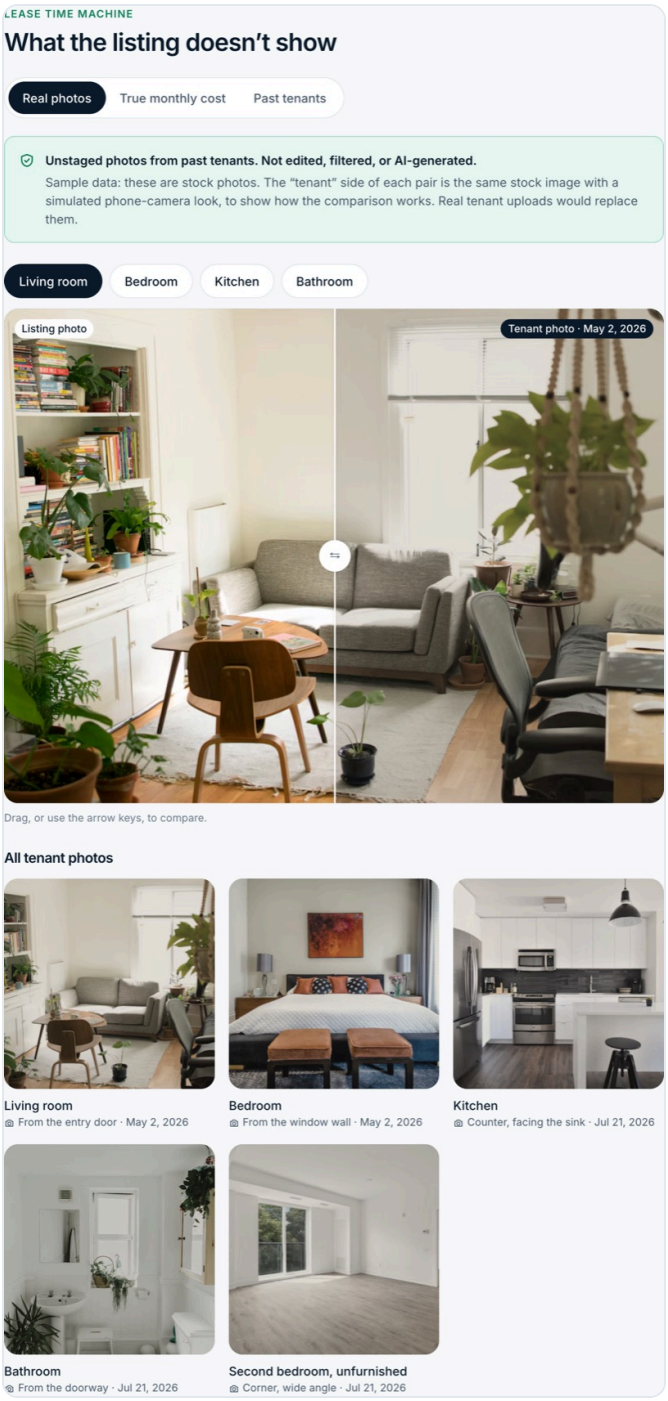


Figure 7. Lease Time Machine, Real photos tab: before/after slider and dated tenant photos, with an honest note that these are illustrative stock pairs.

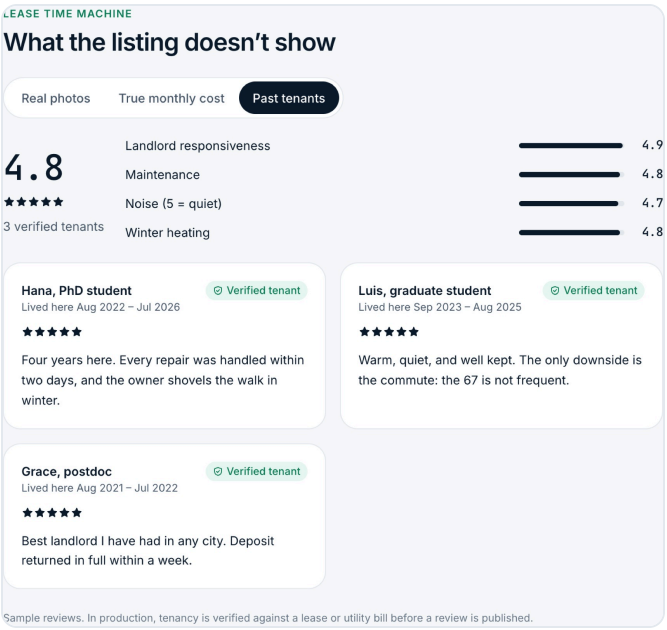


Figure 9. Past tenants tab: category ratings and reviews with a Verified tenant badge and tenancy dates.

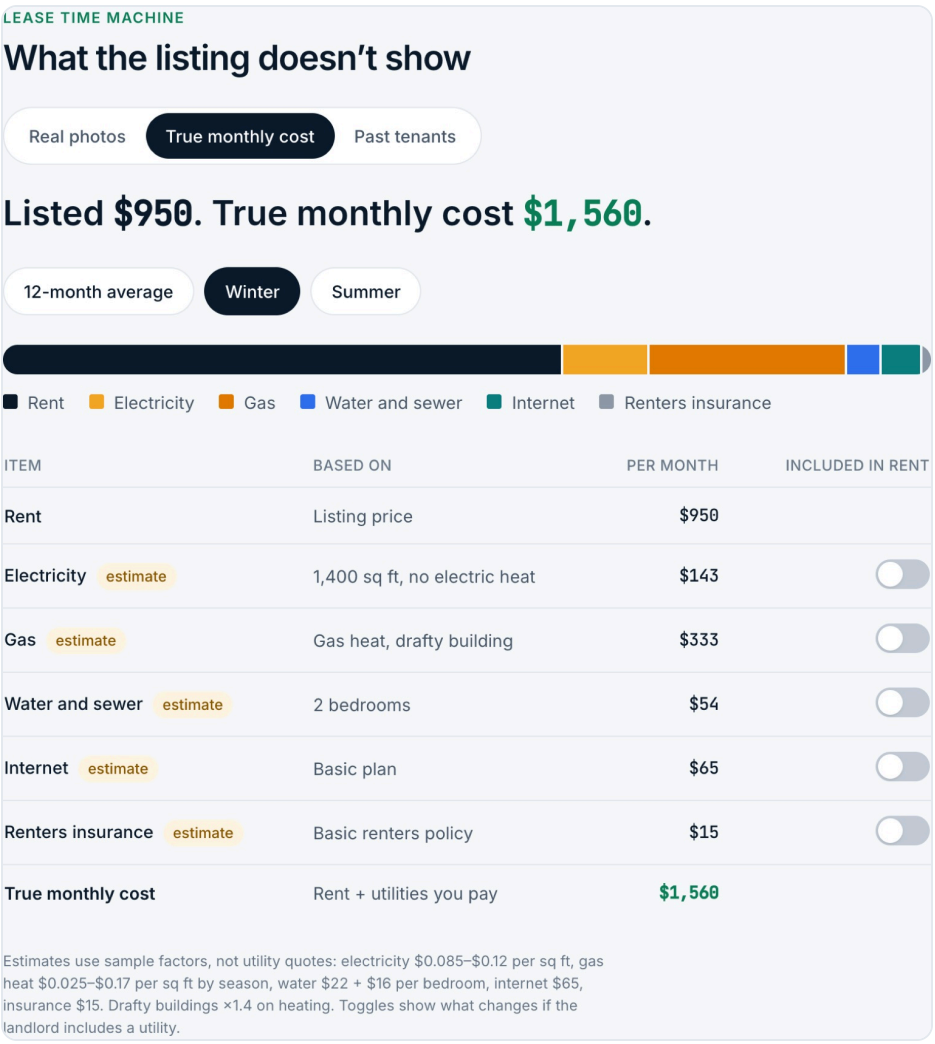


Figure 8. True monthly cost for the cheapest listing, winter: "Listed $950. True monthly cost $1,560." Gas alone is $333 in a drafty house.

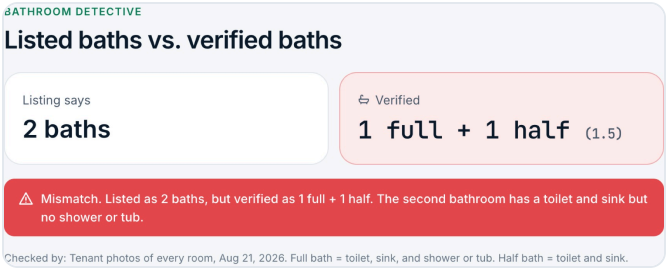


Figure 10. Bathroom Detective: listed 2 baths, verified 1 full + 1 half, red mismatch flag.

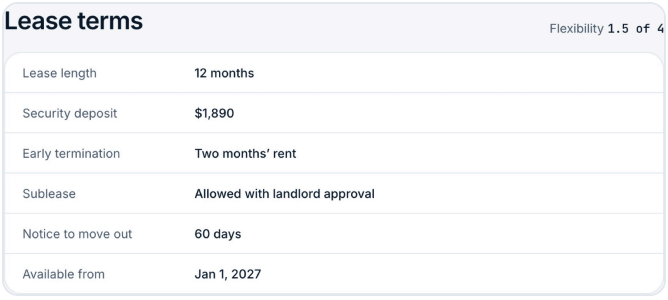


Figure 11. Lease terms with a flexibility score.

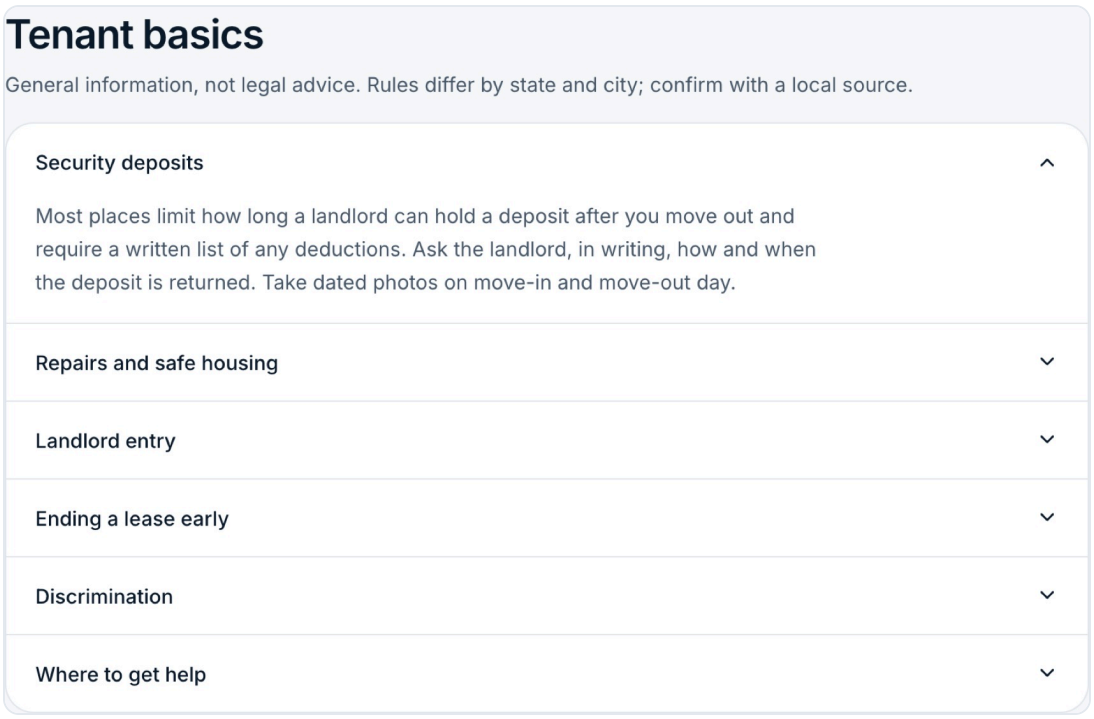


Figure 12. Tenant basics, labeled "general information, not legal advice". No statute numbers.

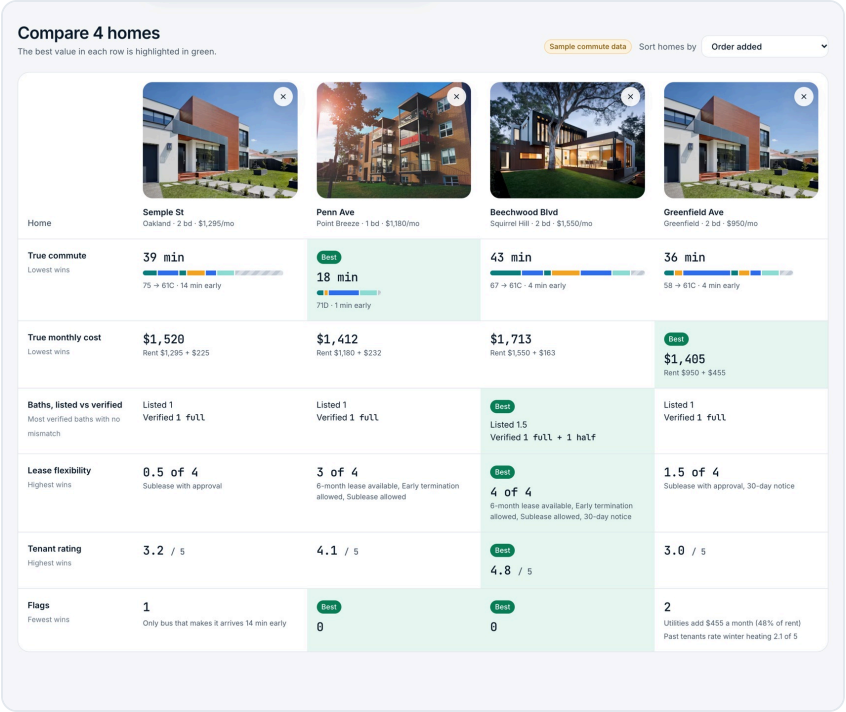


Figure 13. Compare 4 homes. The "Sort homes by" menu reorders them by commute, utilities, true cost, rent, rating, or flags. The best value in each row is green: lowest commute and cost, most verified baths with no mismatch, highest flexibility and rating, fewest flags.

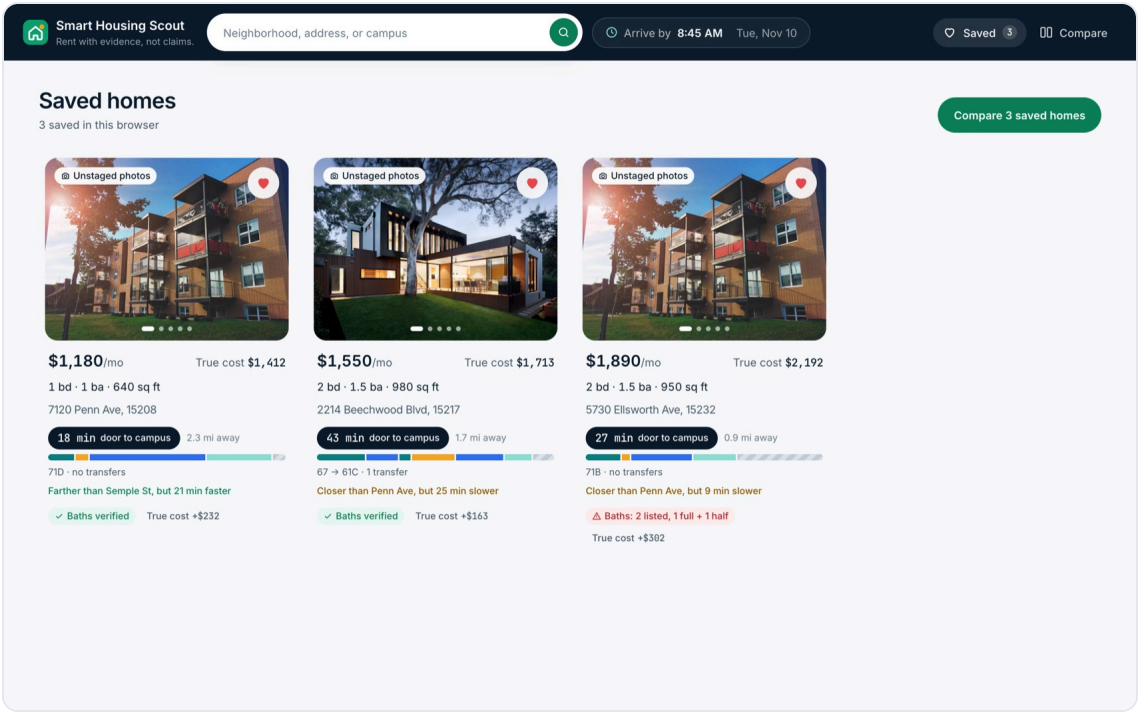


Figure 14. Saved homes, stored in the browser.

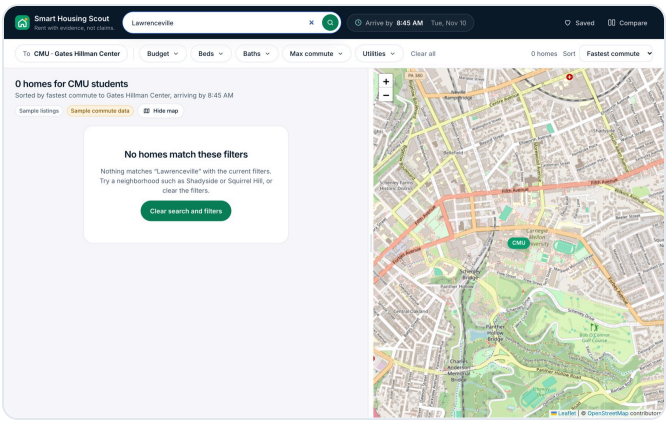


Figure 15. Empty state with a clear next step.

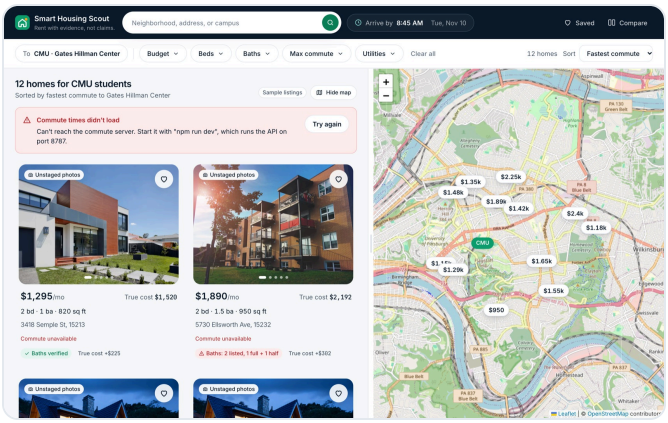


Figure 16. Error state when the commute server is off: what happened, how to fix it, and a Try again button.

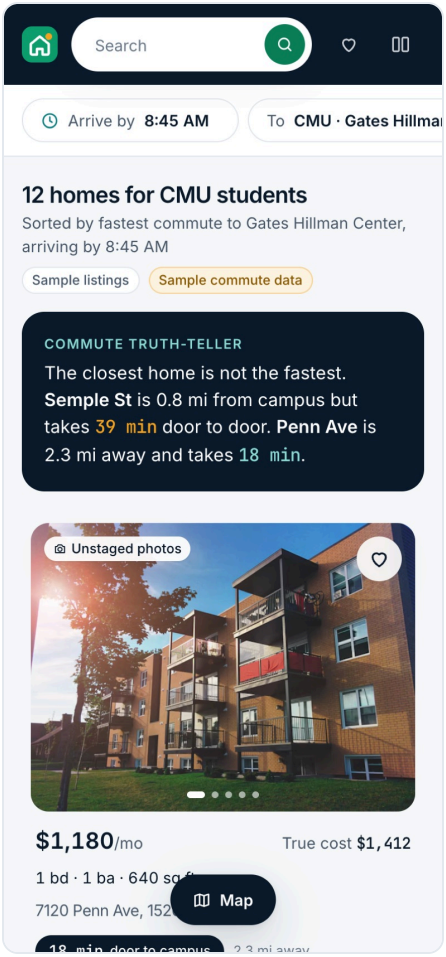


Figure 17. Phone: list view.

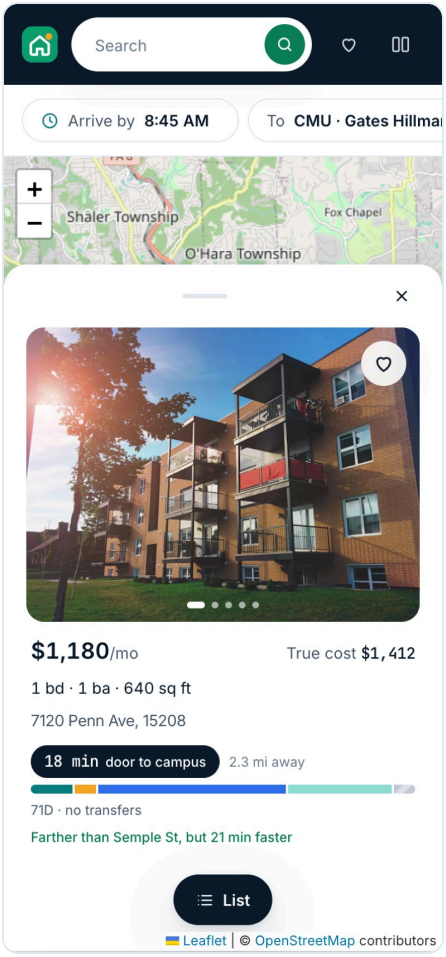


Figure 18. Phone: map view; tapping a pin opens a bottom sheet.

3. Testing verification

Automated tests, run with `npm test` on Nov 9, 2026. All 35 pass. The fixture is a Google Routes API response with one transfer.

AREA	WHAT THE TESTS CHECK	TESTS	RESULT
Route parser	A Google Routes response with one transfer (fixture) becomes walk 5, wait 5, ride 13, transfer walk 2, final walk 4, early arrival 10 = 39 min. Lines 61B and 71A, 1 transfer. Edge cases: missing duration, walk-only route, no route, odd seconds still add up	11	**All pass**
True cost	Hand calculation for a 1,000 sq ft, 2-bedroom, gas-heated home: $92 + $85 + $54 + $65 + $15 = $311 extra. Included utilities, toggles, winter vs summer, drafty buildings	6	**All pass**
Findings	"Closer than Penn Ave, but 21 min slower" and "Farther than Semple St, but 21 min faster"; differences under 3 min ignored; 2 listed vs 1 full + 1 half flagged; 1.5 listed vs 1 full + 1 half accepted; lease flexibility score	8	**All pass**
Sample traps	Closest home is not the fastest; a farther direct bus wins; the best-reviewed home has the longest commute; results repeat exactly	3	**All pass**

AREA	WHAT THE TESTS CHECK	TESTS	RESULT
Time zone	"8:45" becomes the next weekday at 8:45 in Pittsburgh: Monday afternoon to Tuesday, Friday to Monday, late Thursday night stays Thursday, summer daylight time, bad input rejected	7	**All pass**
Total (`npm test` , Nov 9, 2026)		**35**	**35 of 35 pass**

Manual checks in a browser: toggling "Internet included" lowered the Ellsworth Ave total from $2,192 to $2,127 (–$65); hovering a card highlighted its pin; the API rejected `"8.45"` as an arrive-by time with a clear message; a request from another website got no CORS permission.

YOUR TURN. If your instructor wants your own evidence, run `npm run dev` , repeat the five trap checks, and replace or add your own screenshots here.

QUESTION 2C

Code modification (2 points)

Changes made after the first working version, found while testing in a real browser. Pick one to describe.

CHANGE	WHY	HOW
A. Photo labels fixed	The first screenshot showed a house with palm trees as a Pittsburgh listing, and some "bedroom" photos were kitchens. Code tests could not catch this.	Every stock photo was viewed on a contact sheet and re-sorted into living room, kitchen, bedroom, bath, and exterior lists in `shared/listings.ts` . Exteriors were spread so neighboring cards differ.
B. "Farther but faster" compares with the closest home	The Penn Ave card said "Farther than Westminster Pl, but 25 min faster", which compared it with a random slow home. The useful comparison is with the home a renter would pick by distance.	In `commuteInsight()` , slower homes are now sorted by distance, not by time. The card now reads "Farther than Semple St, but 21 min faster".
C. Clear message when the server is off	When the API restarted, the dev proxy answered with an empty 500 error, so the page would have said "Commute server returned 500", which does not tell the user what to do.	In `fetchCommute()` , any 500-level response without a JSON body now shows: "Can't reach the commute server. Start it with npm run dev" (Figure 16).
D. Phone layout	On a 375 px phone the "Arrive by" pill wrapped onto two lines and the search placeholder was cut to "Neighbc". The compare tray covered the map's bottom sheet.	Pills got `whitespace-nowrap` ; the placeholder becomes "Search" on narrow screens; the tray hides in map mode.

WHO DID WHAT

Claude found these problems while testing in the browser, wrote the code changes, and re-ran the tests.

YOUR TURN. Choose one change. Describe what changed, why, and how, in your own words. Say truthfully who typed the code. If you make your own change, describe that one instead.

QUESTION 2D

Summary reflection (3 points)

TRUE EVENTS TO DRAW FROM

Systems approach

- The prompt defined the users, the five test traps, the data types, the build order, and an offline mode before any code was written. The traps worked as a test plan: each one has an expected result that can be checked.
- A review of the prompt before building found the biggest gap: "arrive by 8:45" hides time lost to arriving early. That became the fifth segment and changed how total commute is defined.

Plan compared with the finished project

- Node.js was not installed on the laptop, so Claude downloaded the official version for the session only.
- The traps did not appear on their own. Claude had to adjust the sample bus schedules so the closest home really was slow and the far home really was fastest, and added tests so they stay that way.
- The Google Routes API has no trip start time, which the prompt assumed. The parser works it out from the route duration instead.

Working with the AI

- Most helpful: building a full app with a server, tests, and offline mode from one detailed prompt, and checking the API documentation before writing code.
- Where it needed checking: screenshots caught problems that 35 passing tests missed: wrong photos, a confusing comparison, and phone layout bugs.
- The rate limit worked: the screenshot script itself was blocked after 300 requests a minute, which proved the limit is real.
- Not yet verified: the real Google transit path, because no API key was available.

Next time

- Write expected numbers for each trap in the prompt, not only descriptions.
- Sketch the phone layout too, not only desktop.
- Get an API key early so real data can be tested.

YOUR TURN. Write several paragraphs in your own words: your systems approach, plan vs. result, how you worked with the AI, and what you would do differently.

QUESTION 3

Group review (5 points)

In our group discussion, each of us presented our project and the others tried it. Gautam built a poker game with different game modes, such as a biggest-hand game. His interface was excellent: very clear, with strong colors and contrast, and very interactive. It felt like a full website, not something made just for an assignment, so hats off to him.

Arzu made a bus time checker that tells you when to take the bus. Her interface was very cute, and it removed the boring friction from a daily routine. Tanya made an internship tracker that was simple yet interactive. Max built the backend for the common space project, and I loved how detailed it was, with the punk wall, the desk LED, and the ring countdown.

Tvisha's project was the one I found most helpful for me personally. Her tracker keeps track of what is left in the fridge and helps plan what to make each day. As a student at Carnegie Mellon, deciding what to cook every day without knowing what is left in the fridge is a real struggle, and her tracker handles all of it in a simple, organized way. Right now nothing like it exists unless you ask Claude to manage it for you.

Before you submit

- □ Fill in every YOUR TURN box in your own words.
- □ Check that the statement of who did what is accurate.
- □ Upload to Canvas. Optional: attach the project folder or a link to the code.